TRANSFER OF COHERENCE ALONG TIETZE EQUIVALENCES OF UNDERLYING0-POLYGRAPHS

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Editorial note. Source https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.FSCD.2025.30. © Samuel Mimram and Émile Oleon, CC BY4.0. Complete author natural-language proof and original diagrams retained.

Editorial note. The transfer theorem retains the author coherence hypothesis and complete constructive proof. The separate choice-dependent completeness theorem is not used as a new assumption.

Allowed standard prerequisites (editorial).

- Homotopy type theory, higher inductive realizations, paths/cells and elementary Tietze transformations. Complete author §§1–2.2 and §3 definitions/proofs retained. The axiom of choice used in the separate completeness section is not introduced as an assumption for this transfer theorem.

1 1-polygraphs

Before introducing 1-polygraphs, we first need to present the (quite trivial) 0-dimensional variant. A 0-polygraph P is simply a type whose elements are called 0-generators. A sphere (x, y) in a 0-polygraph P is a pair of elements of P, and we write $\mathrm{S}\,P$ for the type of such objects. If we write $\mathrm{S}^0 = \{0, 1\}$ for the type with two elements, corresponding to the canonical 0-sphere, it can be noted that a sphere in a 0-polygraph P corresponds to an arrow $\mathrm{S}^0 \to P$. Given a 0-polygraph P, we call $\|P\|_{-1}$ the proposition presented by P.

1.1 Internal and external 1-polygraphs

1-polygraphs.

A 1-polygraph P consists of a 0-polygraph $\mathrm{U}\,P$ (the underlying 0-polygraph of P) together with a function $\mathrm{S}\,\mathrm{U}\,P \to \mathcal{U}$, which to any sphere (x, y) in the underlying 0-polygraph associates a type $P(x, y)$ whose elements are called the 1-generators of the 1-polygraph. In other words, 1-polygraphs correspond to type-theoretic directed graphs. A 1-polygraph is discrete when all the types $\mathrm{U}\,P$ and $P(x, y)$ are sets. In the following, a sphere in $\mathrm{U}\,P$ is called a 0-sphere of P.

1-generators.

The elements of $\mathrm{U}\,P$ are called 0-generators or 0-cells of P. We sometimes write $\mathrm{S}_0\,P := \mathrm{S}\,\mathrm{U}\,P$ for the 0-spheres, as well as $P_0 := \mathrm{U}\,P$ and $P_1 := \Sigma((x, y) : \mathrm{S}_0\,P).P(x, y)$ for the type of 0- and 1-generators of P. We also write $\partial^-, \partial^+ : P_1 \to P_0$ for the source and target functions (defined by $\partial^- := \pi \circ \pi$ and $\partial^+ := \pi' \circ \pi$).

1-cells.

Given a 1-polygraph P and a 0-sphere (x, y), we write $P^*(x, y)$ for the type of 1-cells in P from x to y. This family of types is defined inductively as the smallest one such that

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for every 0-cell $x : \mathrm{U}\,P$, we have a 1-cell $\mathrm{id}_x : P^*(x, x)$,

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for every 0-cells $x, y, z : \mathrm{U}\,P$, 1-generator $a : P(x, y)$ and 1-cell $u : P^*(y, z)$, we have a 1-cell $a :: u : P^*(x, z)$,

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for every 0-cells $x, y, z : \mathrm{U}\,P$, 1-generator $a : P(y, x)$ and 1-cell $u : P^*(y, z)$, we have a 1-cell $a ::' u : P^*(x, z)$.

Informally, cells in $P^*(x, y)$ are undirected paths from x to y in the graph corresponding to P (where the two last cases respectively correspond to taking an arrow a forward and backward). In the following, we often write $a : x \to y$ for a 1-generator $a : P(x, y)$ and $u : x \overset{*}{\to} y$ for a 1-cell $u : P^*(x, y)$.

Composition.

Given two 1-cells $u : x \overset{*}{\to} y$ and $v : y \overset{*}{\to} z$, we define their composite $u \cdot v : x \overset{*}{\to} z$ by induction on u by

$\mathrm{id} \cdot v := v$ $(a :: u) \cdot v := a :: (u \cdot v)$ $(a ::' u) \cdot v := a ::' (u \cdot v)$

Lemma 1 ([13, comp0-assoc]).

Composition is associative and unital.

Given a 1-cell $u : x \overset{*}{\to} y$, we define its opposite $u^{\mathrm{op}} : y \overset{*}{\to} x$ by induction on u by

$\mathrm{id}^{\mathrm{op}} := \mathrm{id}$ $(a :: u)^{\mathrm{op}} := a \mathbin{::'} u^{\mathrm{op}}$ $(a ::' u)^{\mathrm{op}} := a \mathbin{::} u^{\mathrm{op}}$ where $a :: u$, which corresponds to appending a to u, is defined by

$a :: \mathrm{id} := a :: \mathrm{id}$ $a :: (b :: u) := b :: (a :: u)$ $a :: (b ::' u) := b ::' (a :: u)$ and $a ::' u$, corresponding to appending a backward to u, is defined similarly.

Spheres.

A sphere s in a 1-polygraph P consists of a 0-sphere (x, y) together with a pair of 1-cells $f, g : x \overset{*}{\to} y$. The underlying 0-sphere (x, y) is noted $\mathrm{U}\,s$. The cell f (resp. g) is called the source (resp. target) of s and noted $\partial^- s$ (resp. $\partial^+ s$), so that the sphere s is the triple $(\mathrm{U}\,s, \partial^- s, \partial^+ s)$. We write $\mathrm{S}\,P$ for type of spheres in a 1-polygraph.

Coherent 1-cells.

In the definition of 1-cell, we think of $a :: -$ and $a ::' -$ as respectively pre-composing by the 1-cell a and its inverse. However, nothing enforces that they are actually inverse operations, i.e. that we have

$a ::' a :: u = u$ $a :: a ::' u = u$ for any 1-generator a and 1-cell u of suitable types. We define the types of coherent 1-cells $P^{\sim}(x, y)$, indexed by 0-spheres (x, y), as the quotient of the types of 1-cells $P^*(x, y)$ under the congruence generated by the above relations. This type is the "right" definition for the 1-cells of P, in the sense that it gives a description of the free groupoid generated by P [9]. However, non coherent 1-cells provides representatives for those which are often simpler to work with. Namely, by definition of coherent 1-cells as a quotient of 1-cells, we have

Lemma 2.

For $x, y : \mathrm{U}\,P$, the canonical map $f : P^*(x, y) \to P^{\sim}(x, y)$ is a surjection, i.e. the type $\| \mathrm{hfib}_f(u) \|_{-1}$ is inhabited for any $u : P^{\sim}(x, y)$.

1.2 Geometric realization

We introduce the "geometric realization" of a 1-polygraph P, which can be thought of as a type-theoretical graph with P_0 as vertices and P_1 as edges. The set presented by P is then defined as the type of connected components of this graph.

Geometric realization.

Given a 1-polygraph P its geometric realization $\overline{P}$ is the higher inductive type generated by the two constructors

$\iota : \mathrm{U}\,P \to \overline{P}$ $\gamma : ((x, y) : \mathrm{S}_0\,P) \to P(x, y) \to \iota x = \iota y$ More abstractly, it corresponds to the pushout on the left or to the coequalizer on the right

$$\begin{array}{ccc} P_1 \times \mathrm{S}^0 & \xrightarrow{f} & P_0 \\ {\scriptstyle \pi'}\downarrow & \ulcorner & \downarrow \\ P_1 & \longrightarrow & \overline{P} \end{array} \qquad\qquad P_1 \underset{\partial^+}{\overset{\partial^-}{\rightrightarrows}} P_0 \longrightarrow \overline{P}$$

where the horizontal arrow f sends an edge $a : x \to y$ and 0 (resp. 1) to x (resp. y), which corresponds to attaching 1-cells to a 0-dimensional cell complex. The associated induction principle is the following.

Proposition 3.

Suppose given an arbitrary type A. Any two functions $f_0 : P_0 \to A$ and $f_1 : (((x, y), a) : P_1) \to f_0(x) = f_0(y)$ induce a unique function $f : \overline{P} \to A$ such that $f \circ \iota = f_0$ and $f \circ \gamma = f_1$. We also have the expected dependent generalization.

By partial evaluation, the map γ induces, for $(x, y) : \mathrm{S}_0\, P$, a map $\gamma_{x,y} : P(x, y) \to \iota x = \iota y$, which extends as a map on cells $\gamma^*_{x,y} : P^*(x, y) \to \iota x = \iota y$ defined by induction by

$\gamma^*_{x,x}(\mathrm{id}) := \mathrm{refl}$ $\gamma^*_{z,y}(a :: u) := \gamma a \cdot \gamma^*_{x,y}(u)$ $\gamma^*_{z,y}(a ::' u) := \gamma a^- \cdot \gamma^*_{x,y}(u)$ Moreover, since this definition is compatible with the relations for coherent 1-cells, for every sphere $(x, y) : \mathrm{S}_0\, P$ we have the following.

Lemma 4.

The map $\gamma^*_{x,y}$ induces a map $\gamma^\sim_{x,y} : P^\sim(x, y) \to \iota x = \iota y$.

This map is an equivalence, so it provides a characterization of the paths in geometric realizations of graphs [13, KvR/1Polygraph/PathPresentation.agda]:

Theorem 5.

The map $\gamma^\sim_{x,y} : P^\sim(x, y) \to \iota x = \iota y$ is an equivalence.

Proof.

A map $\iota x = \iota y \to P^\sim(x, y)$ can be defined using the elimination principle for paths in $\overline{P}$ given in [9, Theorem 3], and both can be shown to be mutually inverse by induction. ◀

Presented set.

The set presented by a 1-polygraph P is the set of connected components of its geometric realization $\|\overline{P}\|_0$. This notion is to be understood as an analogue of a group presentation with generators and relations, but shifted down by one dimension: here, the 0-cells of P are the generators of the presented set, and the 1-cells are the relations.

Example 6.

Consider the polygraph P with $\mathrm{U}\, P = \mathbb{Z}$ and one relation $a_n : n \to n + 2$ for $n \in \mathbb{Z}$. The presented set is $\mathbb{Z}_2$.

We now prove some technical lemmas that will be useful later on.

Lemma 7.

For $x, y : \mathrm{U}\, P$ and $p : |\iota(x)|_0 = |\iota(y)|_0$ in $\|\overline{P}\|_0$, there merely exists a 1-cell $x \overset{*}{\to} y$ in P.

Proof.

By [19, Theorem 7.3.12], the type $|\iota(x)|_0 = |\iota(y)|_0$ coincides with $\|\iota(x) = \iota(y)\|_{-1}$. Since we are eliminating to a proposition, we can suppose given a path $p : \iota(x) = \iota(y)$ which by Theorem 5 induces a coherent cell $\tilde{p} : P^\sim(x, y)$, and we conclude by Lemma 2. ◀ The following shows that we can pick a representative 0-cell in every connected component [13, has-repr]:

Proposition 8.

The type $\Pi(x : \|\overline{P}\|_0).\|\Sigma(\hat{x} : \mathrm{U}\, P).|\iota(\hat{x})|_0 = x\|_{-1}$ is inhabited.

Proof.

Any proposition being a set [19, Lemma 3.3.4], it is enough to construct a map $f : \Pi(x : \overline{P}).\|\Sigma(\hat{x} : \mathrm{U}\,P).\iota(\hat{x}) = x\|_{-1}$ which can be done by using the elimination principle of $\overline{P}$ (Proposition 3). Namely, we define, for $x : \mathrm{U}\,P$, $f(\iota(x)) := |(x, \mathrm{refl}_{\iota(x)})|_{-1}$ and, for $x, y : \mathrm{U}\,P$ and $a : P(x, y)$, the path associated to it by f is canonically defined by the fact that we are eliminating to a proposition. ◀

2 Tietze transformations

Tietze transformations originate from the work of Tietze [18] who defined two transformations on group presentations:

(T0)

adding a definable generator,

(T1)

adding a derivable relation.

The equivalence relation they induce, the Tietze equivalence, characterizes when two presentations give rise to the same group, up to isomorphism. These transformations have then been generalized to a variety of contexts, including the very general one of polygraphs [1, 5] for presentations of (higher) categories, and have as well been shown to provide a interesting notion of weak equivalence [7]. Their fundamental property can be decomposed in two parts:

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correctness: any two presentations which are Tietze equivalent present the same structure,

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completeness: any two presentations of the same structure are Tietze equivalent.

Here, we think of a 1-polygraph P as presenting the set $\|\overline{P}\|_0$, and we define elementary transformations as explained above. We then show that they are correct and complete with respect to the presented set.

2.1 Definition

We define the two following families of Tietze transformations, which transform a given 1-polygraph P into another 1-polygraph, by attaching cells to it:

(T0)

Add a definable 0-generator: given a type X and a function $\partial : X \to \mathrm{U}\,P$, we construct the polygraph $P{\uparrow_0}\partial$ defined, for $(x, y) : \mathrm{S}_0\,P$, by

$$\mathrm{U}(P{\uparrow_0}\partial) := \mathrm{U}\,P \sqcup X \quad (P{\uparrow_0}\partial)(x, y) := \begin{cases} P(x, y) & \text{if } x : \mathrm{U}\,P \text{ and } y : \mathrm{U}\,P \\ x = \partial(y) & \text{if } x : \mathrm{U}\,P \text{ and } y : X \\ \bot & \text{otherwise} \end{cases} \quad \text{(T1)}$$

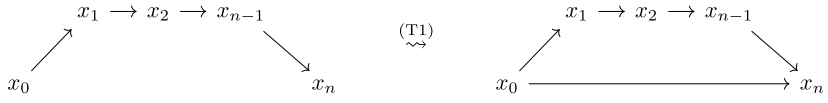
Add a derivable relation: given a function $\partial : \mathrm{S}_0\,P \to \mathcal{U}$ and a family of functions $\partial_{x,y} : \partial(x, y) \to (x \overset{*}{\to} y)$ indexed by 0-spheres $(x, y) : \mathrm{S}_0\,P$, we construct the polygraph $P{\uparrow_1}\partial$ defined, for $(x, y) : \mathrm{S}_0\,P$, by

$\mathrm{U}(P{\uparrow_1}\partial) := \mathrm{U}\,P$ $(P{\uparrow_1}\partial)(x, y) := P(x, y) \sqcup \partial(x, y)$

A transformation (T0) consists in adding, for every element $x : X$, a new 0-generator x to the polygraph together with a 1-generator $\tau_x : \partial x \to x$:

$$y \quad \overset{\text{(T0)}}{\rightsquigarrow} \quad y \xrightarrow{\tau_x} x$$

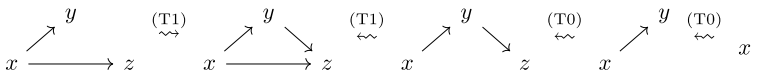
with $y \overset{\text{def}}{=} \partial x$, and τ_x is a notation for the element $\mathrm{refl}_x : (P{\uparrow_0}\partial)(y, x)$. A transformation (T1) consists in adding, for every $x, y : X$ and $a : \partial(x, y)$, a 1-generator $a : x \to y$:



It can be observed that we have a new hole after performing a transformation (T1), which justifies the importance of the set-truncation in the definition of the presented set. Note that the functions $\partial_{x,y}$ are not used in order to define Q in (T1), but they will be necessary to show the correctness of the transformation below. For instance, if we did not require those, we could make any polygraph coherent by taking $\partial(x, y) = \top$ for any $(x, y) : \mathrm{S}_0\, P$.

▶ **Remark 9.**
We could also consider a weaker notion of transformation (T1) where we only require the existence of the 1-cell associated to an added 1-generator, i.e. type the function as $\partial_{x,y} : \partial(x, y) \to \|x \xrightarrow{*} y\|_{-1}$. This variant would in fact be enough for the subsequent developments here.

Example 10.
We have the following series of Tietze transformations:



Under the Grothendieck duality between type families and fibrations [19, Section 4.8], the Tietze transformations (T1) can reformulated as follows (a similar reformulation for (T0) could also be given):

Proposition 11.
Transformations (T1) are equivalent to the following:
(T1')
Given a type $X : \mathcal{U}$ and function $\partial : X \to \Sigma((x, y) : \mathrm{S}_0\, P).P^*(x, y)$, we construct the polygraph $P{\uparrow_1}\partial$ with $\mathrm{U}(P{\uparrow_1}\partial) := \mathrm{U}\, P$ and, for $(x, y) : \mathrm{S}\, P$,

$$(P{\uparrow_1}\partial)(x, y) := P(x, y) \sqcup \mathrm{hfib}_{\pi\circ\partial}(x, y)$$

We define the Tietze equivalence as the equivalence relation generated by (T0) and (T1).

Transformations for 0-polygraphs.
The notion of Tietze transformation generalizes to polygraphs in any dimension. In particular, for 0-polygraphs, it can be formulated as follows. An elementary Tietze transformation of 0-polygraphs consists in
(T0)
Given a 0-polygraph P, a type X and a function $\partial : X \to P$, we consider the 0-polygraph $P{\uparrow_0}\partial := P \sqcup X$.

Two 0-polygraphs are Tietze equivalent when they are related by the smallest equivalence relation identifying P with $P{\uparrow_0}\partial$ for any P and ∂.

Proposition 12.
There exists a Tietze equivalence between two 0-polygraphs P and Q if and only if they present the same proposition, i.e. $\|P\|_{-1} = \|Q\|_{-1}$.

2.2 Correctness

Our aim is now to show the correctness property for Tietze transformations of 1-polygraphs: equivalent 1-polygraphs present the same set.

Theorem 13 (Correctness).
Given two Tietze equivalent 1-polygraphs P and Q, we have $\|\overline{P}\|_0 = \|\overline{Q}\|_0$.

By induction on the equivalence, it is enough to show the result when Q is obtained form P by a Tietze transformation (T0) or (T1), which we do below. Throughout the section, we suppose fixed a 1-polygraph P. We first show that transformations (T0) are correct [13, tietze0-correct]:

Proposition 14.
Given a type X and a function $\partial : X \to \mathrm{U}\, P$, we have $\|\overline{P}\|_0 = \|\overline{P{\uparrow_0}\partial}\|_0$.

Proof.
By definition of $P{\uparrow_0}\partial$ and its geometric realization, we have a coequalizer

$$P_1 \sqcup X \underset{\partial^+ \sqcup \mathrm{id}_X}{\overset{\partial^- \sqcup \partial}{\rightrightarrows}} P_0 \sqcup X \longrightarrow \overline{P{\uparrow_0}\partial}$$

By computing the coequalizer partially on the left component of the coproduct, we find that $\overline{P{\uparrow_0}\partial}$ is the coequalizer of $X \underset{\mathrm{inr}}{\overset{\iota\circ\partial}{\rightrightarrows}} \overline{P} \sqcup X$. From which we easily deduce $\overline{P{\uparrow_0}\partial} = \overline{P}$. ◀ The transformations (T0) actually preserve, not only the presented set, but also the geometric realization. We now show that transformations (T1) are correct [13, tietze1-correct]:

Proposition 15.
Given a type X, a function $\partial : \mathrm{S}_0\, P \to \mathcal{U}$, and a family of functions $\partial_{x,y} : \partial(x,y) \to (x \overset{*}{\to} y)$ indexed by 0-spheres $(x,y) : \mathrm{S}_0\, P$, we have $\|\overline{P}\|_0 = \|\overline{P{\uparrow_1}\partial}\|_0$.

Proof.
By using the induction principle of geometric realization (Proposition 3), we can define a map $s : \overline{P} \to \overline{P{\uparrow_1}\partial}$ by

$s(\iota(x)) := \iota(x)$ $\mathrm{ap}_s(\gamma(a)) := \gamma(\mathrm{inl}(a))$ for $x : P_0$ and $a : x \to y$ in P_1. Conversely, we define a map $r : \overline{P{\uparrow_1}\partial} \to \overline{P}$ by induction by

$r(\iota(x)) := \iota(x)$ $\mathrm{ap}_r(\gamma(\mathrm{inl}(a))) := \gamma(a)$ $\mathrm{ap}_r(\gamma(\mathrm{inr}(b))) := \gamma^*_{x,y}(\partial_{x,y}(b))$ for $x : P_0$, $a : x \to y$ in P_1, and $b : \partial(x,y)$ for some $(x,y) : \mathrm{S}_0\, P$. By induction, they can be shown to form a section-retraction pair, which induces an equivalence on the 0-truncations. ◀

3 Functors between 1-polygraphs

There is a natural notion of morphism between 1-polygraphs. In practice, it is however too restrictive because it requires sending 1-generators to 1-generators, whereas we often need to send 1-generators to 1-cells. In order to allow for this, we then introduce the notion of functor between 1-polygraphs.

3.1 Morphisms

A morphism $F : P \to Q$ between two 1-polygraphs P and Q consists of a function $\mathrm{U}\, F : \mathrm{U}\, P \to \mathrm{U}\, Q$ between 0-generators together with a function

$$F : \Pi((x,y) : \mathrm{S}_0\, P).P(x,y) \to Q(\mathrm{U}\, Fx, \mathrm{U}\, Fy)$$

sending 1-generators to 1-generators. In other words, this corresponds to a morphism of graphs. Those can be composed in the expected way, and the resulting operation is associative and unital.

We write $P \simeq Q$ for the type of isomorphisms between 1-polygraphs. The structure identity principle [19, Section 9.8] ensures that identities correspond to those in the sense that the canonical map $(P = Q) \to (P \simeq Q)$ from identities to isomorphisms is an equivalence.

3.2 Functors

Given two 1-polygraphs P and Q, a functor $F : P \to Q$ consists of a function $\mathrm{U}\, F : \mathrm{U}\, P \to \mathrm{U}\, Q$ together with a function

$$F : \Pi((x,y) : \mathrm{S}_0\, P).P(x,y) \to Q^*(\mathrm{U}\, Fx, \mathrm{U}\, Fy)$$

In the following, for simplicity, we often simply write F instead of $\mathrm{U}\,F$, and leave implicit the first argument of the above function F. Note that any morphism can canonically be seen as functor.

Given a functor $F : P \to Q$, we write

$$F^* : \Pi((x, y) : \mathrm{S}_0\, P).P^*(x, y) \to Q^*(\mathrm{U}\,Fx, \mathrm{U}\,Fy)$$

for the canonical extension of F to 1-cells (as opposed to 1-generators), defined by

$F^*(\mathrm{id}) := \mathrm{id}$ $F^*(a :: u) := Fa :: F^*u$ $F^*(a ::' u) := (Fa)^{\mathrm{op}} :: F^*u$ Given two functors $F : P \to Q$ and $G : Q \to R$, their composite is the functor $F \cdot G : P \to Q$ defined by $\mathrm{U}(F \cdot G) := \mathrm{U}\,G \circ \mathrm{U}\,F$ and $(F \cdot G)(a) := G^*(Fa)$. The composition thus defined is associative and unital. It could alternatively be defined as a Kleisli construction based on the free category monad on 1-polygraphs (details left to the reader).

A functorial equivalence between two 1-polygraphs P and Q consists of two functors $F : P \to Q$ and $G : Q \to P$ together with a family of cells

$\eta_x : x \xrightarrow{*} G(Fx)$ $\varepsilon_y : F(Gy) \xrightarrow{*} y$ for every 0-generators $x : \mathrm{U}\,P$ and $y : \mathrm{U}\,Q$.

Given a polygraph P, the identity functorial equivalence between P and itself is defined by $F = G = \mathrm{Id}_P$ and $\eta_x = \varepsilon_x = \mathrm{id}_x$. Given two equivalences $(F, G, \eta, \varepsilon) : P \to Q$ and $(F', G', \eta', \varepsilon') : Q \to R$, we define their composite as the functorial equivalence between P and R consisting of the functors $F \cdot F'$ and $G' \cdot G$ together with the families of morphisms

$$x \xrightarrow{\eta_x} G'F'x \xrightarrow{G'^*\eta_{F'x}} G'GFF'x$$

$$FF'G'Gy \xrightarrow{F^*\varepsilon_{Gy}} FGy \xrightarrow{\varepsilon_y} y$$

This composition can be shown to be associative and unital. Moreover, any equivalence $(F, G, \eta, \varepsilon) : P \to Q$ admits the equivalence $(G, F, \varepsilon^{\mathrm{op}}, \eta^{\mathrm{op}}) : Q \to P$ as inverse (the opposite is taken pointwise).

3.3 Functions induced by functors

Any functor $F : P \to Q$ induces a function $\overline{F} : \overline{P} \to \overline{Q}$ between the corresponding geometric realizations defined, using the elimination principle of Proposition 3, by

$\overline{F}x := \iota(Fx)$ $\overline{F}a := \gamma^{\sim}(Fa)$ for $x : \mathrm{U}\,P$ and $a : x \to y$ a 1-generator in P, where the map $\gamma^{\sim}$ is defined in Lemma 4. The following is shown in [13, equivalence-pres].

Proposition 23.

By the above construction, any functorial equivalence $(F, G, \eta, \varepsilon) : P \to Q$ induces an equivalence $\|\overline{P}\|_0 \simeq \|\overline{Q}\|_0$.

Proof.

Consider the functions $\|\overline{F}\|_0 : \|\overline{P}\|_0 \to \|\overline{Q}\|_0$ and $\|\overline{G}\|_0 : \|\overline{Q}\|_0 \to \|\overline{P}\|_0$. Given $x : \|\overline{P}\|_0$, our aim is to show that $\|\overline{G}\|_0 \circ \|\overline{F}\|_0(x) = x$. Since this type is a proposition, and thus a set, it is enough to construct a function $\phi : \Pi(x : \overline{P}).(\|\overline{G}\|_0(|\overline{F}(x)|_0) = x)$ which can be done by induction on x. Namely, for $x : \mathrm{U}\,P$, we have $\phi(\iota x) := |\gamma(\eta_x^{\mathrm{op}})|_0$ and, for $a : P(x, y)$, $\phi(\iota a)$ is canonically given by the fact that $\|\overline{P}\|_0$ is a set. We can similarly show that $\|\overline{F}\|_0 \circ \|\overline{G}\|_0(y) = y$ for $y : \|\overline{Q}\|_0$, thus allowing us to conclude. ◀

▶ **Remark 24.**

It is not true that a functorial equivalence between P and Q induces an equivalence $\overline{P} \simeq \overline{Q}$ between the corresponding geometric realizations (as opposed to

presented sets). For instance, consider the polygraph P with only one 0-cell and the polygraph Q with one 0-cell and one 1-cell, whose geometric realizations are respectively the point and the circle: they are functorially equivalent, but their realizations are not.

The above proposition allows us to recover in a "constructive" way the correctness property of Tietze transformations (Theorem 13). Namely, one can show [13, preservation]:

Proposition 25.

Any Tietze equivalence between two 1-polygraphs P and Q induces a functorial equivalence between them.

Proof.

Since functorial equivalences are closed under identities, composition and inverses, it is enough to show that any elementary Tietze transformation induces a functorial equivalence between the corresponding polygraphs.

(T0)

Given a type X and a function $\partial : X \to \mathrm{U}\,P$, we consider the functor $F : P \to P{\uparrow_0}\partial$ which is the canonical inclusion and the functor $G : P{\uparrow_0}\partial \to P$ defined by

$$G(x) := \begin{cases} x & \text{if } x : \mathrm{U}\,P, \\ \partial(x) & \text{if } x : X, \end{cases} \quad G(a : x \to y) := \begin{cases} a :: \mathrm{id} & \text{if } x, y : \mathrm{U}\,P, \\ \mathrm{id}_x & \text{if } x : \mathrm{U}\,P \text{ and } y : X. \end{cases}$$

We take $\eta_x := \mathrm{id}_x$ for $x : \mathrm{U}\,P$, and $\varepsilon_y : F(Gy) \overset{*}{\to} y$ is defined as id_y for $y : \mathrm{U}\,P$ and as the 1-cell corresponding to the canonical 1-generator $\partial y \to y$ given by definition of (T0) for $y : X$.

(T1)

Given a function $\partial : \mathrm{S}_0\,P \to \mathcal{U}$ and functions $\partial_{x,y} : \partial(x, y) \to (x \overset{*}{\to} y)$, we consider the functor $F : P \to P{\uparrow_1}\partial$ which is the canonical inclusion and the functor G, which is the identity on 0-cells and

$$G(a : x \to y) := \begin{cases} a & \text{if } a : P(x, y) \\ \partial_{x,y}(a) & \text{if } a : \partial(x, y) \end{cases}$$

together with η_x and ε_y being both identities for $x : \mathrm{U}\,P$ and $y : \mathrm{U}(P{\uparrow_1}\partial)$.

◀

To sum up, we have introduced three notions of equivalence for 1-polygraphs: Tietze equivalence, functorial equivalence, and equivalence of presented set. The above theorems show that these three notions coincide (Theorems 13, 22, 23, and 25).

4 Transfer of coherence

In a polygraph, cells can be understood as describing the "coherence" of the data, i.e. all the relations it should satisfy. In particular, if we see a 1-polygraph as presenting a proposition (the proposition presented by the underlying 0-polygraph, in the sense of Section 1), the 1-generators should inform us that we can identify any two 0-generators, since the only relevant data is the presence of such a 0-generator. More formally, the presented set should coincide with the presented proposition: $\|\overline{P}\|_0 = \|\overline{\mathrm{U}\,P}\|_{-1}$. We formalize this here and provide conditions for ensuring that we are in such a coherent situation, as well as tools to transfer coherence from a 1-polygraph to another.

4.1 Coherent presentations

A coherence structure on a 1-polygraph is the data of a 1-cell $x \overset{*}{\to} y$ for every 0-sphere $x, y : \mathrm{S}_0\,P$. We say that a 1-polygraph is coherent when it admits a coherence structure. Note that any 0-polygraph P can be extended into a coherent 1-polygraph by adding one 1-generator between every two 0-generators. We are

however interested here in constructing coherent extensions of P which are "small" (or, at least, combinatorially manageable). The techniques developed here should moreover extend to the case of higher-dimensional polygraphs, where the situation is much more involved [1].

4.2 The transfer theorem

Suppose given a 1-polygraph P, a 0-polygraph Q' and a Tietze equivalence of 0-polygraphs between $\mathrm{U}\,P$ and Q' (see Proposition 12). We show that when P is coherent, we can transfer its coherence structure to an extension of Q', which is moreover finite when the one of P is. Namely, we can construct a transferred 1-polygraph Q with $\mathrm{U}\,Q \overset{\text{def}}{=} Q'$ by induction on the Tietze equivalence as follows [13, extension]:

(E+)

if Q' was obtained from $\mathrm{U}\,P$ by an elementary Tietze transformation, i.e. there is a type X and a function $\partial : X \to \mathrm{U}\,P$ such that $Q' = \mathrm{U}\,P \sqcup X$, we define

$$Q(x,y) := \begin{cases} P(x,y) & \text{if } x, y : \mathrm{U}\,P \\ x = \partial(y) & \text{if } x : \mathrm{U}\,P \text{ and } y : X \\ \bot & \text{otherwise} \end{cases}$$

so that there is a Tietze transformation of 1-polygraphs (T0) between P and Q, in the sense that we have $Q := P{\uparrow_0}\partial$,

(E−)

if $\mathrm{U}\,P$ was obtained from Q' by an elementary Tietze transformation, i.e. there is a type X and a function $\partial : X \to Q'$ such that $\mathrm{U}\,P = Q' \sqcup X$, we define

$$Q(x,y) := \Sigma(x', y' : \mathrm{U}\,P \sqcup X).P(x', y') \times (x = \partial'(x')) \times (y = \partial'(y'))$$

where $\partial' : Q' \sqcup X \to Q'$ is defined by

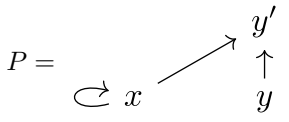
$$\partial'(x) := \begin{cases} x & \text{if } x : Q', \\ \partial(x) & \text{if } x : X, \end{cases}$$

(E∗)

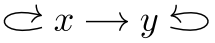
the cases of reflexivity and transitivity are handled by identity and composition.

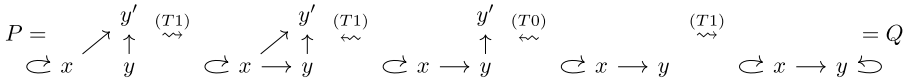
Example 26.

Consider the following discrete polygraphs P and Q':

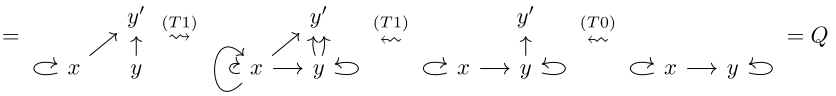
$P =$ 

$Q' =$ 

We have that $\mathrm{U}\,P$ is obtained from Q' by an elementary Tietze transformation of 0-polygraphs with $X = 1$ and $\partial : 1 \to Q'$ defined by $\partial y' = y$, where we write y' for the canonical element of 1. The resulting 1-polygraph Q (obtained by transfer) is thus $Q =$  . The polygraphs P and Q are Tietze equivalent: we have the Tietze transformations



We will show in Theorem 28 that there is always a Tietze equivalence when P is coherent, but the equivalence produced by the proof of the theorem will rather be the following one:



▶ **Remark 27.**
Consider the variant of the previous example where P is the same 1-polygraph with the 1-generator $y \to y'$ removed and Q' is left unchanged. In this case, the resulting polygraph Q is the same as in previous example. However, the two polygraphs are not Tietze equivalent. Namely, we have $\|\overline{P}\|_0 = 2$ and $\|\overline{Q}\|_0 = 1$, and a Tietze equivalence would thus be in contradiction with Theorem 13. Note that the polygraph P is not coherent.

We now show the following transfer theorem which states that a Tietze equivalence between 0-polygraphs can be extended as a Tietze equivalence between 1-polygraphs when one of them is coherent [13, transfer]. Remark 27 illustrates why the coherence hypothesis is required.

Theorem 28 (Transfer theorem).
When P is coherent, the above polygraphs P and Q are Tietze equivalent.

Proof.
As above for the construction of Q, we reason by induction on the Tietze equivalence between $\mathrm{U}\,P$ and Q'. In the case (E+), Q is obtained from P by a Tietze transformation (T0) and we conclude immediately.

The interesting case to handle is (E−), where $\mathrm{U}\,P = Q' \sqcup X$ for some type X and $\partial : X \to Q'$. In this case, we are going to construct a series of 1-polygraphs, which are elementarily Tietze equivalent, thus exhibiting a Tietze equivalence between P and Q, as illustrated in Example 26. It is natural to consider the 1-polygraph R obtained from Q by performing the "same Tietze transformation", but with 1-polygraphs instead of 0-polygraphs, i.e. to consider $R := Q{\uparrow_0}\partial$. Clearly, P and R have the same 0-generators, but not the same 1-generators, and we will show that both are equivalent to a third 1-polygraph S, still with the same 0-cells, and with the union of the 1-generators of P and those of Q as 1-cells. Formally, we thus define the polygraph S, as well as a variant $\tilde{S}$, by

$\mathrm{U}\,S = \mathrm{U}\,\tilde{S} = \mathrm{U}\,P$ $S(x,y) = P(x,y) \sqcup Q(x,y)$ $\tilde{S}(x,y) = Q(x,y) \sqcup P(x,y)$ We claim that we have a series of Tietze transformations

$$P \overset{(T1)}{\rightsquigarrow} S \simeq \tilde{S} \overset{(T1)}{\leftsquigarrow} R \overset{(T0)}{\leftsquigarrow} Q$$

as illustrated in Example 26. Those can be constructed as follows.

■

$P \overset{(T1)}{\rightsquigarrow} S$. The coherence of P provides us, for every 0-sphere (x,y), with a cell $x \overset{*}{\to} y$. This allows us to define a family of functions, indexed by 0-spheres $(x,y) : \mathrm{S}_0\,P$, $\partial_{x,y} : S(x,y) \to (x \overset{*}{\to} y)$, so that we have $S \overset{\text{def}}{=} P{\uparrow_1}\partial$.

■

$S \simeq \tilde{S}$. The polygraphs S and $\tilde{S}$ are clearly isomorphic, in the sense of Section 3.1.

■

$R \overset{(T1)}{\rightsquigarrow} \tilde{S}$. We define a family of functions indexed by 0-spheres $(x,y) : \mathrm{S}_0\,R$, and $\tilde{\partial}_{x,y} : P(x,y) \to (x \overset{*}{\to} y)$ by

$$\tilde{\partial}_{x,y}(a) := \begin{cases} a :: \mathrm{id} & \text{if } x : Q' \text{ and } y : Q' \\ a :: \tau_y :: \mathrm{id} & \text{if } x : Q' \text{ and } y : X \end{cases} \qquad \begin{cases} \tau_x ::' a :: \mathrm{id} & \text{if } x : X \text{ and } y : Q' \\ \tau_x ::' a :: \tau_y :: \mathrm{id} & \text{if } x : X \text{ and } y : X \end{cases}$$

where, for $x : X$, the 1-generator $\tau_x : \partial x \to x$ is canonically given by the definition of R from Q by a Tietze transformation (T0). We then have $\tilde{S} \overset{\text{def}}{=} R\uparrow_1 \tilde{\partial}$.

■

$R \overset{(T0)}{\rightsquigarrow} Q$. The transformation (T0) from Q to R is given by the very definition of R.

◀

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